

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Industry Problem Solving Task

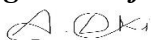
<i>Criteria</i>	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	25%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I A. LOKESH KUMAR (192524157) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1 . Dataset: Supermarket Purchases

Customer Visits/Month Avg Spend (₹) Items Bought

C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

Q2. Dataset: Credit Card Transactions

Transaction Amount (₹) Location Time (hrs)

T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

Q3. Dataset: Telecom Usage

User Call Duration (min) Data Usage (GB) Recharge (₹)

U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

Q4 . Dataset: E-commerce Behavior

User Product Views Clicks Purchases

U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. *(BL3 – 7 Marks)*

Q5. Dataset: Traffic Data

Location Vehicle Count Avg Speed (km/h) Delay (min)

L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. *(BL4 – 7 Marks)*

Answer 1

Understanding the Problem

The supermarket dataset records five customers (C1–C5) through three behavioural attributes: Visits/Month, Average Spend per visit (₹), and Items Bought. A quick review reveals two distinct shopping patterns. C1, C3, and C5 visit frequently (10–12 times a month), buy many items per visit (15–18), but spend a relatively small amount per visit (₹1,800–₹2,100) — consistent with routine, high-frequency grocery shoppers. C2 and C4, in contrast, visit rarely (3–4 times a month), buy very few items (6–8), but spend a large amount per visit (₹5,000–₹6,000) — consistent with occasional, high-ticket bulk or premium shoppers.

Proposed Clustering Approach

Since the number of natural customer types is small and not known with certainty in advance, K-Means clustering is proposed as the primary technique, using Visits/Month, Average Spend, and Items Bought as the three clustering features. Before clustering, all three attributes should be normalized (e.g., Min-Max scaling), because Average Spend is measured in thousands of rupees while Visits and Items Bought are measured in small single or double-digit counts; without normalization, spend would dominate the distance calculation and mask the frequency-based pattern. The Elbow Method (plotting SSE against K) or the Silhouette Score can then be used to confirm the optimal number of clusters — for this dataset the natural elbow occurs at K=2, matching the two behavioural groups visible on inspection.

Applying K=2 to the normalized data produces the following segments: Cluster A – Frequent, High-Volume, Low-Spend Shoppers (C1, C3, C5): customers who visit often and buy many low-cost items per trip, representing the store's routine grocery base. Cluster B – Infrequent, Low-Volume, High-Spend Shoppers (C2, C4): customers who visit rarely but make large-value purchases, representing premium or bulk-buying customers.

How This Supports Targeted Marketing

- Cluster A (frequent shoppers) can be targeted with loyalty-point programs, weekly discount coupons, and subscription-style offers that reward repeat visits, since these customers are highly responsive to frequency-based incentives.
- Cluster B (high-spend, low-frequency shoppers) can be targeted with premium bundle offers, bulk-purchase discounts, and personalized high-value promotions timed around their less frequent visits, since blanket weekly coupons would have little relevance to their shopping pattern.
- Marketing budgets can be allocated more efficiently by matching campaign type and channel (e.g., SMS reminders for frequent shoppers versus curated email offers for infrequent high-value shoppers) to each segment's actual behaviour rather than using a one-size-fits-all campaign.
- New product placement and cross-selling recommendations can be tailored per cluster — smaller, everyday items for Cluster A and premium or bulk items for Cluster B — improving both customer satisfaction and average revenue per visit.

Overall, K-Means clustering on visit frequency, spend, and basket size converts a generic customer list into two actionable, behaviourally distinct segments, allowing the supermarket to design targeted, higher-converting marketing strategies instead of applying uniform promotions across its entire customer base.

Answer 2

Understanding the Problem

The credit card dataset records five transactions (T1–T5) with three attributes: Amount (₹), Location, and Time (hrs, in 24-hour format). Reviewing the values: T1 (₹200, Chennai, 10 hrs), T3 (₹150, Chennai, 11 hrs), and T5 (₹180, Chennai, 12 hrs) are all small, daytime transactions in the same city — consistent with routine, low-risk spending. T2 (₹5,000, Delhi, 2 hrs) and T4 (₹7,000, Mumbai, 1 hr) are both large-value transactions occurring in the very early hours of the morning, in cities different from the customer's usual location (Chennai) — a pattern strongly associated with potentially fraudulent activity.

Proposed Clustering-Based Solution

An unsupervised clustering approach is well suited to fraud detection because fraudulent transactions are rare and often do not have pre-labelled examples to train a supervised classifier on. The proposed solution uses K-Means (or, for a more density-sensitive alternative, DBSCAN) applied to normalized values of Amount, Location (encoded numerically, e.g., via one-hot encoding or by distance from the customer's home city), and Time. Because Amount ranges from ₹150 to ₹7,000 while Time ranges only 1–12, normalization is essential so that Amount does not overwhelm the distance metric.

Applying K-Means with $K=2$ to this normalized data separates the transactions into: Cluster A – Normal Transactions (T1, T3, T5): small amounts, home-city location (Chennai), and daytime hours — representing the customer's typical, expected spending pattern. Cluster B – Anomalous Transactions (T2, T4): large amounts, unfamiliar locations, and unusual (very early morning) hours — representing transactions that deviate sharply from the customer's established behavioural profile. Alternatively, DBSCAN can be used to explicitly treat T2 and T4 as low-density 'noise' points relative to the dense, tightly-packed cluster of normal Chennai transactions, which is a natural fit for fraud detection since DBSCAN does not force every point into a cluster and can flag outliers directly rather than merely assigning them to a 'different' cluster.

How This Detects Unusual or Fraudulent Transactions

- Transactions whose Amount, Location, and Time combination places them far from the dense, low-value, home-city cluster (as with T2 and T4) are automatically flagged as anomalies without requiring pre-labelled fraud examples.
- Using DBSCAN specifically allows the system to identify T2 and T4 as noise/outlier points rather than forcing them into an ill-fitting cluster, which is a more natural representation of rare fraudulent events.
- The combination of unusually high amount, unfamiliar location, and odd transaction time provides multiple independent behavioural signals; a transaction that is anomalous on all three dimensions simultaneously (as T2 and T4 are) receives a much stronger fraud signal than one that is only mildly unusual on a single attribute.
- The clustering model can be re-run continuously on new transaction streams, updating what 'normal' looks like for each customer over time, so genuinely new but legitimate spending patterns (e.g., a planned trip) are not repeatedly misflagged once they become part of the customer's regular behaviour.

In practice, transactions falling into the anomalous cluster (or flagged as DBSCAN noise) would be routed for additional verification — such as an SMS/OTP confirmation to the cardholder or a temporary hold pending manual review — rather than being blocked outright, balancing fraud prevention with customer convenience. This clustering-based approach thus provides a scalable, label-free method for continuously monitoring transaction streams and surfacing potentially fraudulent activity for further investigation.

Answer 3

Understanding the Problem

The Telecom Usage dataset records five users (U1–U5) through three attributes: Call Duration (min), Data Usage (GB), and Recharge (₹). U1 (300, 5, ₹200), U3 (350, 6, ₹220), and U5 (320, 5.5, ₹210) form a group of consistently high-usage, high-recharge customers. U2 (100, 2, ₹100) and U4 (80, 1, ₹90) form a group of consistently low-usage, low-recharge customers. Low usage and low recharge value are classic early indicators of disengagement, since a customer who calls less, consumes less data, and recharges with smaller amounts is likely using the service less overall and may be considering switching providers.

Applying Clustering to Identify Churn-Prone Users

K-Means clustering is applied to the normalized Call Duration, Data Usage, and Recharge values (normalization is necessary since Recharge is measured in tens to hundreds of rupees while Data Usage is measured in single-digit GB units, and without scaling Recharge would dominate the distance calculation). Using $K=2$, informed by the Elbow Method, the algorithm separates users into two clusters: Cluster A – High-Engagement Users (U1, U3, U5): long call durations, high data consumption, and high recharge amounts, indicating active, satisfied, and revenue-generating customers. Cluster B – Low-Engagement, Churn-Prone Users (U2, U4): short call durations, minimal data consumption, and low recharge amounts, indicating users who are disengaging from the service and are at elevated risk of switching to a competitor.

The clustering can be extended over time by re-running it on each user's monthly usage snapshot; a user who gradually moves from Cluster A toward Cluster B across consecutive months provides an even stronger, trend-based churn signal than a single snapshot alone, since it captures a genuine decline in engagement rather than a one-off low-usage month.

Improving Customer Retention Strategies

- Users identified in the low-engagement cluster (U2, U4) can be proactively targeted with retention offers — such as discounted recharge plans, bonus data, or loyalty rewards — before they churn completely, rather than waiting until after they have already left.
- Because the cluster is based on multiple usage dimensions (calls, data, recharge) rather than a single metric, the retention team gets a more holistic and reliable churn signal than relying on recharge amount alone, which could be misleading for a user on a long-term prepaid plan.
- High-engagement users (U1, U3, U5) can be simultaneously targeted with loyalty and upsell offers (e.g., premium data packs) to further increase their value, since clustering reveals not just who is at risk but also who represents the strongest growth opportunity.
- Retention campaigns can be tailored to the specific behavioural profile of the churn-prone cluster — for example, offering a data-usage boost to a segment whose data consumption dropped sharply, rather than a generic discount that may not address the actual reason for disengagement.

Overall, clustering transforms raw telecom usage logs into an actionable early-warning system: by continuously monitoring which cluster each user falls into, the telecom operator can identify churn risk before a formal cancellation request is made, and can direct retention resources specifically toward the users who are both likely to churn and valuable enough to be worth retaining.

Answer 4

Understanding the Problem

The E-commerce Behavior dataset records five users (U1–U5) through three engagement attributes: Product Views, Clicks, and Purchases. U1 (50, 20, 5), U3 (60, 25, 6), and U5 (55, 22, 5) show consistently high engagement across all three metrics. U2 (30, 10, 2) and U4 (20, 8, 1) show consistently low engagement across all three metrics. Importantly, within each group the ratio between views, clicks, and purchases is fairly similar, suggesting that these are two genuinely different types of shoppers rather than simply noisy variations of one underlying pattern.

Suggested Clustering Technique

K-Means clustering is suggested for this dataset, applied to the (optionally normalized) Product Views, Clicks, and Purchases values. Since all three attributes are on relatively comparable small-integer scales here, normalization has a smaller effect than in the earlier datasets, but is still good practice to ensure Purchases (which has the smallest numeric range) is not under-weighted relative to Views. Using the Elbow Method or Silhouette Score to validate K, K=2 is well supported for this dataset, producing: Cluster A – Highly Engaged Users (U1, U3, U5): high product views, high clicks, and multiple purchases, representing active, high-converting shoppers. Cluster B – Low-Engagement Users (U2, U4): low product views, low clicks, and minimal purchases, representing casual browsers or users at risk of disengaging from the platform.

As an alternative or complementary technique, Hierarchical clustering could be used first on a sample of the user base to visually confirm this two-group structure via a dendrogram, before deploying the faster K-Means algorithm at full scale across the platform's entire user base for ongoing, real-time segmentation.

How This Improves Recommendation Systems

- Recommendations can be tailored to each cluster's engagement level: highly engaged users (Cluster A) can be shown more advanced or complementary product recommendations (e.g., cross-sell and upsell suggestions) since they have already demonstrated strong purchase intent, while low-engagement users (Cluster B) can be shown simpler, broadly appealing, or discount-led recommendations designed to convert their first meaningful purchase.
- Within each cluster, the recommendation engine can prioritize collaborative filtering using only other users from the same behavioural cluster, since users with similar view-click-purchase patterns are more likely to share genuine product preferences than the platform's user base as a whole, improving recommendation relevance.
- New or infrequent users can be quickly assigned to the nearest existing cluster based on their initial browsing behaviour, allowing the recommendation system to apply that cluster's proven strategy immediately rather than waiting to accumulate enough individual interaction history to build a personalized profile from scratch (addressing the 'cold start' problem).
- Tracking how a user's engagement metrics shift over time, and consequently which cluster they belong to, allows the recommendation system to adapt — for example, escalating personalization and offers as a Cluster B user's engagement begins to rise toward Cluster A's profile.

In summary, clustering users by Product Views, Clicks, and Purchases converts a large, undifferentiated user base into behaviourally coherent groups, allowing the recommendation system to apply group-appropriate strategies, improve relevance for users with sparse individual history, and ultimately increase both engagement and conversion rates across the platform.

Answer 5

Understanding the Problem

The Traffic Data dataset records five locations (L1–L5) through three attributes: Vehicle Count, Average Speed (km/h), and Delay (min). L2 (500, 15, 25) and L4 (600, 10, 30) show high vehicle counts, low average speeds, and long delays — classic indicators of heavy congestion. L1 (200, 30, 10), L3 (220, 28, 12), and L5 (210, 32, 9) show lower vehicle counts, higher average speeds, and short delays — indicating free-flowing traffic conditions. The inverse relationship between vehicle count and speed, and the direct relationship between vehicle count and delay, is consistent with real-world traffic congestion dynamics.

Proposed Clustering-Based Solution

K-Means clustering is proposed, applied to normalized Vehicle Count, Average Speed, and Delay values (normalization is important since Vehicle Count ranges in the hundreds while Speed and Delay range in the tens, and without scaling Vehicle Count would dominate the distance calculation). Using the Elbow Method to validate K, K=2 is well supported for this dataset, producing: Cluster A – Congested Zones (L2, L4): high vehicle counts, low average speeds, and long delays, representing locations experiencing significant traffic congestion. Cluster B – Free-Flowing Zones (L1, L3, L5): moderate vehicle counts, higher average speeds, and short delays, representing locations with healthy traffic flow.

For a city-wide deployment with many more monitored locations, DBSCAN would also be a strong alternative, since it can automatically identify congestion 'hotspots' as dense clusters of high-vehicle-count, low-speed readings without requiring the number of congestion zones to be specified in advance, and can flag genuinely unusual locations (e.g., an isolated accident-related slowdown) as noise rather than forcing them into either the congested or free-flowing group.

How This Helps in Traffic Management

- Traffic authorities can prioritize infrastructure interventions — such as signal timing adjustments, additional lanes, or rerouting — specifically at locations identified in the Congested Zones cluster (L2, L4), rather than spreading limited resources evenly across all monitored locations.
- Real-time or near-real-time clustering of live sensor data allows the system to detect when a previously free-flowing location (Cluster B) begins shifting toward the congested cluster's profile, enabling proactive measures (e.g., dynamic signal retiming or early public alerts) before congestion becomes severe.
- Clustering results can feed directly into public-facing navigation and traffic apps, allowing commuters to be automatically routed away from locations in the Congested Zones cluster and toward Free-Flowing Zones, reducing overall network congestion through distributed route choices.
- Longer-term clustering trends (e.g., which locations consistently fall into the congested cluster during specific times of day) can inform urban planning decisions, such as where new roads, public transit routes, or park-and-ride facilities would have the greatest impact on relieving chronic congestion.

In summary, clustering Vehicle Count, Average Speed, and Delay converts raw sensor readings from individual locations into an actionable map of congestion versus free-flowing zones, allowing traffic management authorities to target interventions precisely, respond proactively

to emerging congestion, and support both real-time navigation systems and longer-term urban planning decisions with data-driven evidence.